

SUMMARY

Software Engineer with **2+ years of production experience** building backend systems, AI-powered applications and automation using **Python, Go, Java/Spring Boot and AWS**. Experienced with **AI agents, LangChain, AutoGen, LLM APIs, prompt engineering, tool/function calling and agent orchestration**. Strong backend foundation across REST APIs, microservices, asynchronous workflows, databases, Redis, Docker, Kubernetes, CI/CD, automated testing, observability and production debugging. Experienced with **Cursor, Claude and GitHub Copilot** for AI-assisted coding, debugging, refactoring, code review and AI-output validation.

TECHNICAL SKILLS

AI & Agent Engineering: AI Agents, LLM Applications, LangChain, AutoGen, LLM APIs, Prompt Engineering, Tool Calling, Agent Orchestration, Multi-Agent Systems, AI Automation, RAG — **Languages:** Python, Go, Java 8+, SQL, JavaScript, TypeScript, Rust, C++ — **Backend:** FastAPI, Django, Django REST Framework, Spring Boot, Node.js, Express.js, REST APIs, Microservices, API Design, Service Orchestration, Background Processing, Asynchronous Workflows — **Databases:** PostgreSQL, TimescaleDB, Redis, MongoDB, MySQL, Oracle, Firebase Firestore — **Cloud & Infrastructure:** AWS, EC2, S3, Docker, Kubernetes, GitHub Actions, Jenkins, Maven, Nginx, CI/CD — **Testing:** JUnit, Mockito, pytest, Unit Testing, Integration Testing, Functional Testing, API Testing, Security Testing, Mocking — **Engineering:** Application Architecture, API Design, DSA, Design Patterns, Functional Requirements, Non-Functional Requirements, Performance Optimization, Authentication, Authorization, RBAC, Production Troubleshooting — **Observability:** Grafana, Plutono, Splunk, Monitoring, Production Observability, APM, Time-Series Analytics — **AI-Assisted Development:** Cursor, Claude, GitHub Copilot, AI-Assisted Coding, Code Review, AI Output Validation — **Development:** Git, GitHub, Version Control, Jira, Confluence, Agile, Scrum, Scripting

EXPERIENCE

Full Stack Engineer

Feb 2026 - Present

Zwilling Labs (IIT Bombay)

- Owned backend development across **8 enterprise deployments**, building **Go/Rust services, Python automation and REST APIs** for high-volume, high-performance industrial telemetry workloads on a customized Plutono/Grafana stack.
- Designed **application architecture, API contracts, service orchestration and end-to-end data flows** across scanner, frontend, backend and database layers; built PostgreSQL/TimescaleDB schemas and indexes for production telemetry workloads.
- Built and integrated **production AI-agent and LLM-powered automation workflows** using **LangChain, AutoGen and LLM APIs**, integrating prompt engineering, tool/function calling and external REST APIs for multi-step workflow automation and agent orchestration.
- Owned the **Hitachi deployment end-to-end**, supporting **150+ machines** with operator, supervisor and admin workflows, task allocation, machine assignment and role-based business rules.
- Reduced automated PDF report generation from **hours to under 2 minutes** using Python-based reporting workflows and Chromium rendering for production reporting.
- Deployed services using **Docker/Kubernetes** with Services, ConfigMaps, Secrets and scaling; built **GitHub Actions CI/CD** workflows across private-cloud, edge and on-premise environments.
- Performed **unit, integration, functional, API and security testing** across boundary and failure scenarios; used production monitoring, observability and **Splunk** for troubleshooting and performance optimization.
- Used **Cursor, Claude and GitHub Copilot** for AI-assisted development, code generation, debugging, refactoring and PR review; validated AI-generated code through testing and review.

Software Engineer (Backend)

Jan 2025 - Jan 2026

Vervelo

- Built backend systems across **4+ client projects** using **Python/Django and Java/Spring Boot**, developing REST APIs for healthcare workflows, notifications, messaging, reports and payment integrations.
- Reduced API response time by approximately **30%** through SQL optimization, indexing and **Redis caching**; implemented authentication, authorization, RBAC and background processing.
- Built Python automation workflows integrating backend services, external APIs and structured data flows; implemented asynchronous processing for production workflows.
- Performed **unit, integration, functional and API testing** using JUnit/pytest; worked with Maven, Jenkins, Git/GitHub and Dockerized services across CI/CD workflows.
- Deployed backend services on **AWS EC2** and performed production root-cause analysis across application, API and database layers.

Web Developer

Feb 2024 - Dec 2024

Thinkitive Technologies, Pune

- Built **React enterprise dashboards**, integrated REST APIs and reusable components; performed API/integration testing and debugging.

RESEARCH EXPERIENCE

Research Project - Lind-Wasm

2026 - Present

*with Justin Cappos, New York University*Contributing to **Lind-Wasm** using **Rust/Wasmtime** for software fault isolation, system-call mediation, virtual memory, signals and WebAssembly.

Research Project - CodeRevival

2026 - Present

with Dennis Shasha, New York University

Developing automated workflows to revive legacy GitHub repositories through dependency, test-configuration and package-version updates.

SELECTED PRODUCT BUILDS

RelayAI - Multi-Agent AI Automation Platform

[GitHub](#)

- Built a **multi-agent AI automation platform** using **Python, FastAPI, LangChain, CrewAI and LLM APIs**, orchestrating specialized agents with tool/function calling, RAG, vector search, external REST APIs, structured outputs, asynchronous workflows, task delegation, retries and failure handling.

Mesh - Multi-Agency SaaS Operating System

[Demo](#)

- Built and deployed a configurable **multi-agency SaaS** used by **200+ users**; implemented REST APIs, relational models, authentication, RBAC, permissions, workflow states and business-process automation.

Morph - Multi-Restaurant Digital Menu SaaS

[Demo](#)

- Built a multi-restaurant SaaS with digital menus, QR ordering and feedback workflows; developed **WhatsApp Business/Meta API** integration for messaging and automated customer communication.

EDUCATION

Fr. Conceicao Rodrigues College of Engineering, Mumbai

Nov 2021 - Mar 2025

B.E. Computer Science (Honors in Blockchain)